



Events and Weak Narrativity

Chapters 3 and 4 argue for the dependence of a viable narrative practice hypothesis on weak narrativity as component to basic cognitive processing, and Chap. 5 introduced prominent models of cognition based upon prediction, a ‘default’ mode of cognition, and embodiment. This chapter argues that weak narrativity exists in low-level cognition. Weak narrativity is characteristic of how we process our experiences as events.

6.1 TIME PROCESSING: PRESENCE AND SEGMENTATION

Cognitive systems are complex, and hard distinctions between low-level and higher order systems are problematic. Research that demonstrates how integral they are to one another provides evidence of the irreducibility of cognition to a set of discrete functional systems. Theories such as the Default Mode Network, Global Workspace, and others, introduced in Chap. 5, provide further evidence that cognitive processes are heterogeneous but deeply intertwined. As empirical research continues to reveal how cognition works, our understanding becomes more complex and diverse, rather than simplified into a unified theory.

Our experiences of time reveal complexity in subjective experience. Time is not experienced as unitary, but as chunked into sometimes clearly defined and sometimes ambiguous parts and intensities. The passage of experience involves duration as well as a succession of instantaneous

moments. With the alternation between attending closely to the perceptual experiences of the present environs and 'default mode' cognition (in which attention to the immediate environment is reduced to a sentinel role while internal cognitive processes are more intensive), experience is segmented.

William James's famous description of consciousness offers an early account of this 'chunkiness.' The stream metaphor was, for James, a way to contest the orthodoxy of late nineteenth-century science. Radical in his time, James rejects the idea that experience consists of a chain of individual units:

The traditional psychology talks like one who should say a river consists of nothing but pailsful, spoonsful, quartpotsful, barrelsful, and other moulded forms of water. Even were the pails and the pots all actually standing in the stream, still between them the free water would continue to flow. It is just this free water of consciousness that psychologists resolutely overlook. Every definite image in the mind is steeped and dyed in the free water that flows round it. With it goes the sense of its relations, near and remote, the dying echo of whence it came to us, the dawning sense of whither it is to lead. The significance, the value, of the image is all in this halo or penumbra that surrounds and escorts it,—or rather that is fused into one with it and has become bone of its bone and flesh of its flesh; leaving it, it is true, an image of the same thing it was before, but making it an image of that thing newly taken and freshly understood. (p. 255)

James' description verges on several paradoxes: the 'free water' of consciousness is simultaneously a 'halo' or 'penumbra' of particular images; it 'surrounds and escorts,' but also (or 'rather') is 'fused' with the image. The image is thus self-identical and particular to its momentary manifestation. James' metaphor of 'steeping' and 'dyeing' resolves these semi-paradoxes. Any image of consciousness is suffused with its 'near and remote' relations, and thus with its past and future (anticipated) changes. James thus suggests that our experiences are in part determined by complex *relational dynamism*. The subjective nature of our experiences, and the particularity of their phenomenology, results from near and distant self-relations as well as relations with 'free' features of consciousness.

James' description illuminates how temporality figures into cognition.¹ Time is not experienced in itself, but as a feature of ongoing consciousness and of particular experiences. Also, time is not experienced as unitary. It is 'chunked' into sometimes clearly defined and sometimes ambiguous parts

and intensities. The parts result from the nature of experience, which involves durations as well as a succession of instantaneous moments. The intensities reflect the nature of those experiences. When attention is highly focused on a succession of percepts, time is experienced differently from when attention to percepts is less focused and the subject is engaged in rumination. Through these parts and varying intensities, experience is segmented.

For example, imagine doing a boring, repetitive task like sweeping a large empty room (say, an empty rectangular dance ballroom). The sweeper might at times forget his bodily motion and feel relatively detached from his surroundings. Imagine that he enters a state of reverie as he recalls a particular evening from his youth. The way he experiences present time would be modulated by the memory, including a possible experience of a gap in present time experience. Then, imagine the sweeper's reveries are interrupted when he stubs his toe on a nail sticking out of the flooring. The initial moments of pain return the sweeper's attention to the present, and that return may include an intensified focus that makes a few seconds seem longer than they would in a period in which no intense perceptions are experienced.

In this simple example there are several factors governing the experience of time. First, the sweeper's own moods and thoughts have durations. These reflect the sweeper's particular attributes as well as larger contexts, such as stressors in the sweeper's life. Also, the sweeper's circadian rhythms and metabolic processes create temporally extended feelings of fatigue and wakefulness, hunger or satiety, and the like. Also, external cues, such as the sweeper's progress through his task, the light coming in through the windows, and unanticipated sounds and interruptions shape temporal experience, as would communicative cues such as the time displayed on his mobile phone.

Taken together, these various processes comprise temporal experience or experiences. Are they fully synthesized into a singular 'temporal experience'? It seems more likely that the sweeper has competing 'temporal experiences,' and that shifts between these comprise the phenomenology of temporal experience. So, the changes described above between states of attention are segmented during a baseline of messy, shifting factors that comprise the sweeper's sense of time. This complexity can occur without the sweeper intentionally training his attention upon time, or upon a particular goal-directed task.

The addition of goal-directed intentionality brings further complexity to this situation. Imagine, for example, that the sweeper's supervisor has recently criticized his work in a particular area of the room, and threatened him with consequences if he doesn't improve. So, when the sweeper comes upon this area, he may train particular focus on this area, which may have additional consequences for how he experiences the time spent in this area versus other parts of the room, and his memory of the work might include an outsized weighting of this part of the work in relation to the amount of time the work took to complete.

As our example suggests, experiences of time are multiply determined by a variety of cognitive processes. These processes exist in competitive 'push-pull' relations among one another and are also shaped by our baseline capacities. Carlos Montemayor explains that our time sense is determined in part by 'cognitive clocks' with capacity limitations of about three seconds (see also Wittmann, pp. 45–50). These clocks mark sensory experiences, which facilitate the coordination of representations of time with "other representations, such as spatial representations and even episodic-like memories" (*Minding Time*, p. xiii, p. xii).

Yet we also experience, via anticipation and working memory, a sense of ongoing continuity that lasts beyond these three second intervals. Our experience involves, as Henri Bergson noted, a feeling of duration. That is, instead of taking each moment independently, they are experienced as a continuity. As Marc Wittmann describes the matter,

What do we mean when we speak of the "present moment," the "instant," or the "now"? Experience has presence. When we see, hear, and feel something, we do so at this very moment: right now. What was only just present belongs to the past in the next moment. In this way, a flow of felt time emerges: we anticipate an event, then we experience it, and shortly thereafter it lies in the past. We experience duration. Conversely, one can also say that we are constantly living in the moment always exactly now. To be sure, events and our experience of them change in status: first they are expected in the future, then they are experienced fleetingly, and finally they remain only in our memory. All the same, our experiences are tied to the moment. A familiar saying holds that we live from one moment to the next—that we move from one instant to the following one, as it were. However, it might be more appropriate to say that our conscious experience is determined by a constant being-present through which events pass: from futurity, which is not yet fixed, to what is experienced in the moment, and then onward, into the past. (p. 44)

Wittmann's model of 'felt time' describes a steady-state phenomenological character for the present, or as he puts it, a 'constant being-present' (p. xiii). Simultaneously, experience is segmented into events which have some existence for us in the future, in the present, and in the past.

This duality is paradoxical in the sense that dynamism, represented in a stream of different events, seems to be coupled with a static state ('constant being-present'). If this stasis is generated by a particular orientation toward present events, and present events are natural phenomena that our orientation disposes us to anticipate and remember, the paradox is apparently resolved.

However, in this description the cognitive work of anticipating and remembering must be considered. There is a continual simulation of future events, which requires residual grappling with events in the past. The seemingly constant state of presence, in other words, is an active processing state. So, the paradox is resolved not by an orientation/event dyad, but by understanding how feelings of continuity emerge in an active processing state.

The example of the sweeper tracks shifts in attention between present sensory inputs (including the feelings associated with the work, such as changes wrought by the onset of fatigue; stubbing his toe; internal physical feelings (such as the onset of hunger), and the condition of the area subject to his boss's criticism) and internal reflections (including planning thoughts associated with completing the work, a memory of an evening from youth, replaying the confrontation with his boss, and thoughts about a potential partner he's recently met). How can these quite different foci of attention be felt as part of continuous processing? The answer is partly embedded in the question: feelings of presence are highly associated with a phenomenology of modulating attention.

In addition, the 'passage' of events from future to present to past describes a sequential, serial chain that requires elaboration in four areas. First, does the event experience have a phenomenology that differs from the phenomenology of the feeling of presence? As a corollary, if the phenomenology of event experience subsumes the feeling of presence, how does the sense of continuity-in-presence manifest? Second, what defines events? Are the boundaries natural or constructed? As they unfold, are event boundaries containers of their constituents or defined by them? Third, can events contain events? As a corollary, what are the typical time signatures of events—that is, how long at minimum and maximum can they be, and how much 'content' can they contain, to be a singular event?

Fourth, if events are anticipated, experienced, and remembered, how do events from these different time stages influence one another? To specify, do other events prime us to anticipate, experience, and remember events in particular ways? As a corollary, how do schema develop from experiences with events, and how are schema deployed in making sense of events?

In what follows, I'll explore these questions regarding attention and events by considering recent research that illuminates their function. Along the way, I'll highlight the emergence in event segmentation and episodic memory processes of dynamics fundamental to the definition of weak narrativity described in chapters 3 and 4. Narrative processing is ongoing meaning-directed generation and processing of emergent temporal structures consisting of related value-dynamic sequences involving value-dynamic selected agents, mediated by socio-cultural contexts. Subsequently, I'll argue that weak narrativity provides a coherent means of characterizing these processes within an overdetermined 'messy ecology' of emergent temporal and semantic experiences. In this way, weak narrativity is helpful for making sense of the phenomenology of event experiences.

6.2 ATTENTION AND EVENTS

Attention is fundamental to time experiences. It is simultaneously responsive to environmental conditions and subject to intentional control. Webb and Graziano (2015) promote a model of attention—Attention Schema Theory—that acknowledges this complexity: “top-down versus bottom-up distinction[s] in attention research ... [are] flawed ... because some effects do not fall neatly into either category” (p. 4).

Attention Schema Theory offers a partial answer to the questions raised above: How does modulating attention relate to an ongoing feeling of presence? For Webb and Graziano, the key is that attention is greater than, and subsumes, awareness. Awareness is dynamic and selective but continuous: “awareness is an internal model of attention useful for the control of attention” (p. 5). This internal model creates a sense of “subjective awareness. *A creature with an attention schema should be a creature that concludes it is aware*” (p. 6).

The feeling Wittmann describes of ‘constant being-present’ is the phenomenology of awareness. As Webb and Graziano put it, the observing perspective produced by our attention schema is experienced as “an amorphous, nonphysical, internal power, an ability to know, to experience, and to respond, a roving mental focus—the essence of covert attention without the underpinning details” (p. 43).

The awareness model, then, comprises a selection from the attentional field that reflects particular priorities. The model provides subjective continuity to our experiences. How these selections are made and coordinated in dynamic, unfolding time, and how they are formulated into units that we can recall and ultimately use to guide our future attention shapes event construction.

Events, as Thomas Shipley (2008) suggests, “appear to be a fundamental unit of experience” (p. 5). Events are experienced in the temporal present, and they are refined and repurposed in episodic memory. Events can be imagined, during which the outcomes and meanings of events may seem manifest in the unfolding of the event. Even during present, actual events, subjects often predict outcomes and meanings and apply them to ongoing dynamics.²

The degree to which events are autonomous from and precede human attention is subject to debate. For theorists subscribing to naturalism, event boundaries occur naturally rather than only as part of meaning-laden human intercourse. In this view events typically manifest in patterns of dynamic changes occurring in a single object or agent. Take the following example of a natural event: a tree falls in the forest. The observer observes the event by perceiving the increased volume of the creaking of the wood and the motion of the tree relative to its surroundings. She tracks the fall until it is complete and the tree is settled into a new position on the forest floor (i.e., indicating the end of the event: the dynamism has stopped). Once the settling has ceased, the subject either attends to the consequences of this event (animals are scattering, more sunlight enters the grove from the direction formerly occupied by the tree, etc.), or simply switches attention to some other event or feature of reality.³ In this case, the event is perceived via multimodal dynamic patterns of onset, development, and cessation. For example, the sounds that contribute to event definition include the onset and increased volume of the cracking wood, and the cessation of sound related to the tree.

In an event such as this, cognition may involve ongoing construction of predictions that guide and even correspond to perception. However, perception is initially reactive—the event occurs in the world, and the subject’s cognition tries to make sense of it.

Constructivists offer a different portrait of events. In their perspective, the mind actively selects and organizes perceptions into meaningful, and ultimately memorable, event units that in turn help guide future cognitive activity. That is, cognition involves constructing events as units of experiential meaning. For the constructivist, the mind organizes experience

based on predictions, conceptual knowledge, and prior experiences. Perception is not merely receptive—instead, it is cognitively structured from the outset. Events are not purely naturalistic phenomena but rather are a key part of how we define and structure experience. An event is never ‘natural’; instead, it is always constructed into meaningful scenes, units, etc. via cognition. As Schwartz argues, “events are what we make of them” (p. 54).

To illustrate this difference, consider our experience of events depicted in film. Generally, new events are experienced by the film viewer as inextricable from meanings established regarding agents and other events in the film. For example, a film director might create the following scene for a spy thriller: a secret agent (spy #1) obtains a forbidden document and seeks to leave a city to deliver it to his handlers. An opposing secret agent (spy #2) seeks to stop spy #1 from escaping. Spy #1 enters a large urban crowd and attempts to blend in. Spy #2 tracks him through costume changes and rapid switches in modes of transport and direction. Spy #1 creates obstacles behind him; spy #2 eludes these obstacles. Eventually the options for locating spy #1 dwindle and spy #2 either captures him or is eluded. Cut scene.

In this analogy, we see several key differences from the naturalistic event of a tree falling in the forest. First, the actors in the scene come laden with semantic meanings—identities—as ‘spy 1’ and ‘spy 2’ (established in some manner prior to the scene, for the sake of simplicity). Second, their dynamic changes are meaning-laden, rather than natural. Spy 1 is more or less in danger of capture, and spy 2 is more or less close to success in capturing her target. In this instance, dynamic changes do not have self-contained meanings (such as the cracking noises that signal the tree falling and the cessation of noises that indicate the end of the event). Instead, dynamic changes create states that indicate a higher likelihood of other dynamic changes (simply, when spy 1 hides behind the door when spy 2 enters the room, spy 1’s capture and both of their deaths seem likely imminent possibilities).

Third, rather than singular dynamism, we have coordination of two dynamisms (changes in spy 1 and spy 2’s conditions, both independently and relative to one another). These dynamisms depend heavily on understanding the goals of each agent, and from a higher order synthesis of the situation that makes the event’s meanings (e.g., danger to the republic or populace on some level). Taken together, each agent’s condition is evaluated in terms of the relative proximity of the other (for spy 1, close proximity of spy 2 changes his value to more endangered; for spy 2, close

proximity changes her value to more likely to achieve her goal by apprehending spy 1).

Boundary delimitation and selection are foundational to these dynamics. In this example, the two agents might be selected out of a crowd that includes hundreds of other people. The film director cues this selection through camera syntax, such as centering each agent in the frame, close ups, repetition, tracking each agent's eyes, and the like. Through these cinematic codes, they become figures highlighted against the 'ground' of the urban crowd.

Finally, selection is inseparable from context-based affective valuations (and perhaps other forms of emotional valuation as well). In this case, our understanding of each agent's relation to social and political hegemonies has primed an affective relation for the audience. For example, perhaps spy 1 is a foreign mole threatening a beloved homeland; or perhaps spy 1 is exposing deep corruption among the powers that be. Spy 2 is selected in relation to spy 1. If we value spy 1 as 'bad' or 'good' (or, in all likelihood, something a bit more complex), spy 2's motives are similarly shaded. In turn, our valuation of spy 2 depends upon her relation to her quest in relation to our valuation of it (and so on...). This all depends on our evaluation of the more general conditions that prevail in the filmic universe. So, if 'selection' involves a distinguishing of 'figure' from 'ground,' this selection is strongly relational and intertwined with more abstract contextual valuations (derived both from the director's choices and from our own value systems).

As may seem obvious, both scenarios—the tree falling and the spy thriller—are quite distant from how we understand events during daily experience. The tree falling scenario may not be an 'event' in any meaningful sense at all—it is merely a 'happening' in the world to which the subject may or may not attend, which may or may not have meaning, dependent on factors such as its proximity to the subject, its uniqueness, and its relevance relative to other factors within the subject's perceptual frame or internal ruminations. The second is a generic, highly designed narrative example that has little to do with how we typically experience the world. Obviously, no filmmaker cues and guides our perceptions of the world through camera work, mise en scène, editing, and other techniques.

Actually experienced events typically seem more dependent upon human valuations than the tree falling, and less designed than the film—we have in these examples relative extremes that might support naturalistic and constructivist accounts. One significant difference is that the example of the tree falling exists without obvious reference to a context outside of

its own dynamism (and in naturalistic views, any attempt to find meaning in the event—say, signs of rot in the forest, an omen of evil, or an instance of the ordinary course of nature—is applied retroactively, rather than part of the event itself). By contrast, the film scene is decipherable as an event, or chain of events, because of prior representations of the agents and their relations to the social order, as well as, potentially, iconic representational cues in the agents' features, comportment, or style that signifies particular meanings. In addition, the two 'events' are quite different in temporal extension and complexity, and the film scene may be best understood as a narrativized chain of individual events that form a chain of meaning.

Yet if we consider these hypothetical scenarios on a continuum within which events fall, their similarities may be as instructive as their differences. Each, obviously, is bounded by time markers: a beginning and an end. Each has a meaningful sequence of internal changes, with meaning determined by progress toward the ground (for the tree), and by relative and related increases in proximities to danger (for spy 1) and success (for spy 2). For each the sequence is predicated on changes in stability: from a relatively stable state, into increased instability and dynamism, and back to a relatively stable, though different, state. Moreover, the second stable state can be reassessed to form a *propositional*, and thus abstractable, account of the event as a whole: the tree has fallen, spy 1 has been captured or has escaped (or better, the chase has failed or succeeded). This propositional accounting is open to additional meanings. From a cognitive perspective, the events' passages entail both an initial attention change (to the tree or spies) and a closing attentional change (either to other stimuli or to rumination on the implication of the events).

Given these differences and similarities, how do we define and characterize events? One definition has some of the structures we associate with closed-form narratives: "a segment of time at a given location that is perceived by an observer to have a beginning and an end" (Zacks and Tversky 2001, cited in Zacks et al. 2007, pp. 1–2). Other definitions of events are narrower, and involve the notion of goals. For example, in "Event Memory in Infancy and Early Childhood," Patricia Bauer notes that she uses "a definition of 'event' borrowed from Katherine Nelson (2003)": events "involve people in purposeful activities, and acting on objects and interacting with each other to achieve some result" (p. 11). Because purposeful activity unfolds over time, it has a beginning, a middle, and an end. Because the actions in events (if defined in terms of goals) are oriented

toward a goal or result, there are often constraints on the order in which they unfold: actions preparatory to an outcome must occur before it in time. This definition excludes simple physical transformations such as fluttering because they do not involve actors engaged in purposeful activity. In contrast, the definition includes chains of activities as complex as world wars. Importantly for present purposes, it also includes the activities in which individuals engage as they move through a typical day.

Event definitions tend to be centered around the experiencing subject as a goal-driven actor. However, subjects can experience events as observers, and they can imagine events. These events require a modulation of what we ordinarily mean by ‘goal-directedness,’ but all three subject-event relation paradigms involve experience parsing in terms of value as well as identity. Value provides the bridge between identification and goals, and it provides a common mechanism for the three paradigms. Value is a synthetic concept (bringing together affect, action-orientation, object-identification, association, and identification) that is nevertheless not secondary to experience, or post-experiential. Instead, it is integral to online, real-time sensemaking (for a description of valuation as it relates to affect, see Chap. 7, esp. Sect. 7.4 to the end). It is attractive as a concept in part because it includes modal representation while also being translatable to amodal representation (see Barsalou 1999b, 2008).

What components characterize events? Despite differences, including their natural or cognitive origins, the examples we’ve considered of events (the tree falling and the spy scene) have a repetitive structure—a sequence of disruption of a prior order and resolution (that is, a return to a relatively static state, though not identical to the initial state⁴). One way to parse this in terms of processes, as seen earlier, is that an event’s borders involve a transition from an explicit or implicit equilibrar state to a disequilibrar state, and then to a (different) equilibrar state. To enact these transitions, there is a state shift from a state of steady or increasing disruption, followed by a state of steady or increasing order.

Given this pattern characteristic of events, what comprises ‘equilibrium’ and ‘disequilibrium?’ That is, what indicates an ‘orderly’ versus a ‘disrupted’ state, and how does ‘orderly’ state 1 relate to ‘orderly’ state 2? Further, how do both orderly states relate to the disrupted state?

Alternations between equilibria and disequilibria may differ based upon whether they are unexpected or expected. Unexpected perceptions, such as the pain one feels as one stubs one’s toe, create an immediate

disequilibrium that resolves into a new equilibrium condition as the pain subsides. Expected perceptions, such as the firing of a rocket engine as one pushes the ignition button on a model rocket, are accompanied by a mediating sense of anticipation. Witness, in this case, the phenomenology of a rocket engine not firing after one has pushed the appropriate button. The non-happening may be experienced as an event precisely because it contravened expectations.

Shifts from disequilibrium to equilibrium conditions may typically trigger shifts from perceptual to reflective processing. For example, after unexpected events, like the stubbed toe, intrude into perception, they are followed by reflective processing during which the individual seeks identification of the source of pain and reflects upon the causes (which may have to do with immediate environmental or internal circumstances, or with long-term evaluations of self and context). Expected events have a more complex integration of reflective processing, as perceptual inputs are highly selected and viewed in terms of anticipated values. Reflective processing also occurs, as with unexpected events, immediately following the perception.

The question of whether events are natural or constructed is further complicated by their revision over time. Via thought, subjects frequently define, refine, or revise the boundaries of an event. However, in large part events come ‘ready made’—that is, their temporal character and boundaries are not the product of conscious construction, but rather seem to exist as part of the character of experience.⁵ Also, event processing requires ‘binding’ multimodal perceptual inputs, as well as bottom-up, sensorimotor and top-down, schematic or anticipatory inputs. As Shipley notes, “To the extent that separate aspects of an event are processed and combined, we must consider how event-feature binding is achieved. In an event, the space-time behavior of objects must be combined” (p. 21). Empirical evidence suggests that the binding of the natural occurs via mediation by models, which are learned, constructed forms. But it is hard to reconcile what Shipley describes—combined space-time behavior of object (and agents)—with the taxonomic, associative form we typically associate with schema. The next section considers two models of event processing which address questions of event boundary definition. Then, I argue for a way to make sense of the deployment of schema in experience based on emergent weakly narrative forms: sequenced, dynamic clusters of valuations parsed into temporal strands coincident with action to fact shifts.

6.3 IN-EVENT PROCESSING

One theory of event processing with considerable empirical support rests on the idea that the natural/constructed dichotomy is mediated through the nature of perception. In ecological and ideomotor approaches, perception is not passive reception of stimuli, as the classical model would have it, but involves active picking up of information from the environment. This active perceiving is inseparable from action and action-planning.

In Hommel's Theory of Event Coding (TEC), events are encoded in event files that bind action plans to perceptual inputs, which in turn mediate the perception of future events.

Action plans are instantiations of goals that, in turn, develop through time via the subject's experience of her ability to affect the world. TEC "assumes that humans are able to register features of (self- or other-produced) events and to integrate the codes of these features into event files" (Hommel, 2004).⁶ These event files are representations in the sense that "they stand for events that do not need to be present" (Hommel 2019, p. 87). According to Memelink and Hommel,

TEC claims that perception and action features are coded in a common format and assumes that perception, attention, intention, and action share and operate on a common representational domain (Prinz 1990). This notion implies that *perceiving an object and acting upon that object is essentially the same process and involves the same network of represented features*. Accordingly, perception may influence action, and vice versa. (2017, p. 250, emphasis added)

TEC provides a model for the mechanisms of event encoding, but as Hommel notes, the boundary definition and internal mechanisms of events are more elusive: "How is an event defined? This question is notoriously difficult with regard to perceptual objects and it is even more difficult regarding events. When and under which (exogenous and endogenous) circumstances will a percept and an action plan be integrated into the same event file, and when into separate files?" (Hommel 2004, p. 499).

A partial answer lies in what Memelink and Hommel term 'intentional weighting'—that is, our intentions to act prime perception of particular features of the environment integral to intended actions (Memelink and Hommel 2013). Their theorized intentional weighting mechanism "operates on all feature dimensions that are involved in discriminating between

task relevant and task irrelevant stimuli and/or responses,” and it weights particular categories of stimulus rather than particular stimulus values (so, color as a category, rather than red) (2013, p. 252).

The intention-weighting principle has a similar explanatory benefit to the notion of ‘affordances’ in earlier ecological views, but leaves room for greater flexibility in perception, as particular stimuli are selected for attention, but not necessarily processed in terms of their task-relevance. This is a valuation mechanism, integral to internal event definition. Nevertheless, it does not seem to offer a nuanced theory of *value dynamism* within events (by value dynamism, I mean changes in the values of agents, objects, and other features within the event—see the definition of weak narrativity above).⁷

Event definition may be partly explained by combining Hommel’s ‘intentional weighting’ with value dynamism. In this model, events are internally defined by coordinated tracking of the dynamism attending to actors that are intentionally weighted. Importantly, this reverses the traditional relationship between an event’s ‘container’ boundaries (the beginning and ending) and changes in the value of actors and other features *within* the event. Instead, changes in value are boundary defining. That is, the ‘container’ is made by its ‘internal’ elements.⁸ Recent findings about event duration support an inside-out model of event emergence. Faber and Gennari (2015) argue that the experience of event duration depends upon the similarities and differences among subevents.

To envision the difference, consider this alteration of a traditional view of boundaries as containers of events (i.e., defining events by first delimiting their outer boundaries (their beginnings and ends)). To stay close to the container metaphor, imagine the formation of a droplet on a countertop. When liquid hits the countertop, the liquid spreads until the combination of its inertia, surface tension, and the shape and slope of the countertop arrest it into a static form (along with other factors). The internal elements, combined with the environment and a dynamic action create the shape and size of the droplet. In event terms, the beginning and end emerge from these dynamics.⁹

So, we seek an explanation of how event boundaries (like beginnings and endings) are related to changes in the values of the elements within the event. Moreover, we require explanation of the more complex phenomenon of relational valuation dynamism. That is, how do the relationships between agents, objects, and other features change, and how does this affect the perceived value of each (see the spy thriller example for an illustration of this). That is, how do ‘weightings’ of particular stimuli depend upon proximity and relation to other stimuli?

Overall, the ‘task-orientation’ of TEC, as of ideomotor theories in general, may seem limiting. As the next chapter explores, affective valuation may play a significant role in perceptual weighting, as may accidental factors such as prior familiarity, strangeness, beauty, ‘pop-out’ qualities (e.g., a strong odor in relation to other environmental stimuli). In addition, analogical similarity to highly valued remembered stimuli may shape selection and weighting. Moshe Bar theorizes that the brain continually generates analogies or “associations as the building blocks of predictions...that prime our subsequent perception, cognition and action” (2007, p. 280).¹⁰

Simply put, not everything we experience as events is derived from top-down, subjective goals. Even if it is possible to reduce each of these ‘weighting’ qualities, and other forms of valuation, to ‘action-orientation,’ such an explanation would entail a wide range of valuations that reflect prior action-oriented encodings rather than the simpler mechanism of task-oriented selection Hommel and colleagues envision. To reduce all to action orientation, while perhaps explanatory of a subject’s experiential history, does not seem promising as a means of explaining the totality of top-down weighting occurring in perceptual presence. If the model requires that we see in perceptual presence a deep history of weightings that can be abundant, potentially contradictory, and still ultimately grounded in (initial/learned) action-orientation, then we begin to approach a view of perceptual selection with a rich phenomenology.¹¹ However, even in such a model we are (a) only experiencing action-orientation in a distal manner; and (b) not explaining how the boundaries of and internal dynamics of events are processed. Therefore, it seems likely that the task-orientation dynamics described in TEC play a significant and foundational role in selection but are not fully explanatory of how we understand events.

While goals have been well-documented to shape selection, our varied experiences suggest the need for an expanded view of the causes of selection. Consider the famous ‘Gorilla and Basketball’ experiment, in which subjects were instructed to count the number of passes successfully completed by basketball players in a brief video. In the course of the video, a figure in a gorilla suit crosses in front of the camera, yet many of the participants did not report seeing this figure. This experiment has been used to illustrate how task-orientation creates selective attention (see Simons and Chabris 1999 for a popular treatment of the ‘inattention blindness’ phenomenon).

We may not see the gorilla while we are counting the number of times the ball is passed between players when we are instructed to focus solely on a singular task, but we should remember that this was a lab experiment, and the subjects were instructed to focus upon counting passes of the basketball. Our ordinary mode of experiencing is not cued in this narrowly goal-directed manner. Instead, we derive the task at hand from a muddier experience, and we manage task-orientation alongside other less structured ‘sentinel’ and exploratory attentional tasks (see Buckner et al. 2008, p. 19 for ‘The Sentinel Hypothesis’ regarding the Default Mode Network (also Chap. 5, Sect. 5.2.2)). Our event processing involves parsing what we are doing, but also what is happening outside of our doing, and this more passive information gathering seems evolutionarily vital.

In addition, once we’ve theorized the mechanisms of event boundary setting, we will look at their potential effects. Event borders, as they emerge, may contribute to in-event processing, event segmentation, episodic memory encoding and retrieval of events, and likely in the feedback through which these processes affect one another, as well as the ways other aspects of sensemaking and higher order cognition affect and are affected by them. Theories of how events are defined vary in their view of representations of the present, of the past, and of similar events; of their views of time perception and its relation to event definition; and of how high- and low-level processing are related (Buckner 2010, pp. 4–7).

6.4 PREDICTION FAILURE AND EVENT BOUNDARIES

So, what determines when an event starts and stops, where it occurs and where (proximally) it is understood to not occur, who and what are involved, and what dynamics reside within an event (versus what dynamics are extra-event)? TEC theorizes that our experiences of events beginning and ending correspond to the opening and closing of event files. Yet, as noted above, this correspondence does not solve the problem of how event parameters are defined. Zacks and colleagues have pioneered some of the most insightful research into this question. Events and their boundaries are, like narratives,¹² ‘fuzzy’ (2007, p. 2). They focus on an ‘everyday’ notion of events typically characterized by certain types (‘meals,’ e.g.), goal-directedness, animate agents, and short duration (seconds to tens of minutes) (2007, p. 2). Their Event Segmentation Theory (EST) seeks to explain “how human observers segment continuous activity into discrete events” (p. 2). The basic mechanism derives from the Predictive Processing framework:

One central proposal of the EST model is that event segmentation is a spontaneous outcome of ongoing perception and arises at points where perceptual predictions begin to fail or change. This change in perceptual predictions is considered to occur at many different levels simultaneously and is a consequence of the failure of a currently active set of predictions, which are rapidly dysfacilitated and superseded by a more accurate set of predictions. The period of epoch between two such consecutive changes current mental models is an *event* (see Zacks et al. 2007, for a review of the extensive body of data demonstrating high sensitivity to event boundaries). (*Understanding Events*, p. 591)

EST hypothesizes that event boundaries are defined by changes in ongoing predictive processing.¹³ In the course of a subject experiencing an event, an event model guides predictions that shape perception (Zacks et al. 2007, p. 3). The event model is an evolving, temporary framework that synthesizes information from event schemata stored in long-term memory with information from prediction errors. Event schemata store “prior knowledge about goals and plans and prior statistical learning” (p. 11).

As an event proceeds, predictions are relatively accurate. However, as it ends, prediction error increases, causing a reset of the event model. When such a reset occurs, an event boundary is set. This model does not, however, view event processing as a serial chain of single, discrete events—EST hypothesizes that event segmentation occurs “simultaneously on a multitude of timescales, spanning a few seconds to tens of minutes” (p. 5). As such it differs from TEC, which seems focused on the short end of this time range.

Events are typically segmented in relation to high-level information like goals and plans, contexts, similarities to previously experienced events, and low-level physical stimuli, especially physical changes. Event perception and segmentation, as noted in the summary of TEC, integrate multimodal information. Richmond and Zacks (2017) note that event models “nonlinearly” transform sensory and motor information into “re-representation[s] of the world” that facilitate “efficient learning and robust extrapolation” (p. 3).

So, EST offers an alternating pattern of relative stability followed by disruption and, subsequently a return to relative stability (marking the instantiation of a new, currently accurate set of predictions). This pattern of “stability punctuated by phasic transitions ... segment[s] ongoing experience into a succession of events” (p. 18). In so doing, it manifests the

patterns of equilibrium-disequilibrium-equilibrium described at the beginning of this chapter, but in a quite different manner. Events, as described earlier, are structured in equilibrium-disequilibrium-equilibrium cycles *internal* to the (narrative) event. So, where for EST the cycle characterizes the sometimes tumultuous transitions *between* events (and thus seems reminiscent of Thomas Kuhn's well-known model of paradigm shifts in scientific discovery), understanding events also requires understanding the governing internal logic of events and the processes that produce them and make them available for further cognition.

Speer, Zacks, and Reynolds offer this account of how readers parse narratives into events:

As incoming information about the current event becomes less consistent with the current mental model, readers are more likely to update the current model to more accurately represent the state of the world (Gernsbacher, 1990; Reynolds, Zacks, & Braver, in press; van Dijk & Kintsch, 1983; Zacks, Speer, Swallow, Braver, & Reynolds, 2007; Zwaan & Radvansky, 1998). Therefore, readers may explicitly perceive the points where characters, goals, and other elements change as event boundaries because these changes drive the updating of readers' mental models at those points. (p. 454)

The proximity of narrative reading to event segmentation provides some support for the notion that we are particularly attuned to narrative and apply the same cognitive processes to making sense of experience that we employ when reading a novel. Indeed, Speer et al. argue that "this process of segmentation [used in detecting the structure of narrated activities] is part of a larger, modality-independent system involved in the comprehension of everyday activities" (p. 454).

In contrast to my characterization, though, Speer et al. describe internal segmentation on the basis of change. However, such a view creates a new issue to explain: the ongoing grasping together of the change-points into a meaningful unity. That is to say, to account for our experience of events as meaningful context-related sequences, we need to account for how the markers laid down by failures in predictive modeling are processed together into ongoing syntheses.

EST hypothesizes, as an implication of its view that event segmentation occurs at multiple timescales simultaneously, that there are principles for hierarchically arranging timescales into coarse and fine-grained predictions nested at multiple levels (p. 19). For example, Zacks considers "getting

ready for work” a candidate coarse-grained event, with “brushing teeth” a nested, but still coarse event consisting of fine-grained events like “putting toothpaste on a toothbrush” (p. 19). In addition, Richmond, Zacks et al. describe *discontinuous* events, such as “building a sculpture,” and connects EST to Kubovy’s “concurrent strands” framework, in which “one might have different representational strands for the many facets of one’s life, including home, work/school, etc.” (p. 19).¹⁴ In EST, people “selectively attend to one timescale” (p. 8).¹⁵

This view of events extends EST well beyond the temporal limits hypothesized in TEC, as well as beyond the direct action plan-percept coupling fundamental to TEC. It therefore may be commensurate with theories of accreting autobiographical narratives considered in some humanistic accounts of narrative and human experience. Before linking into these higher order theories, however, let’s consider the linkages between narrativity and event definition and segmentation inchoate in the psychological research.

To do so, we’ll examine research regarding episodic memories, which Zacks and colleagues see as linked to event processing. How episodic memories are formed out of experience, in what state they exist, and how and to what ends they are deployed comprise a crucial part of cognitive processing that combines affective valuations with sensemaking operations such as recognition of changes and selection and boundary delimitation of place, time, objects, and agents. As such, they seem valuable candidate phenomena for weak narrativity, and their function in relation both to higher order processes and to mediating predictive processing may thereby reveal whether there is anything characteristically narrative about low-level cognition, especially event processing.

6.5 MEMORY AND EVENT PROCESSING

While in classical views of cognition memory areas of the brain function as a storehouse for remembered events, more recent research emphasizes the role memory plays in processing present experiences. This newer emphasis reflects the predictive processing framework. EST sees deep affinities between event segmentation and episodic memories, and between controlling actions and understanding others’ actions (Richmond and Zacks 2017).

Much of this change in views has been prompted by research on the hippocampus, which is typically understood in the popular imagination as

the seat of memory. In the last 20 years, researchers have found the hippocampus to be highly active in tasks not associated with reflecting on one's past.¹⁶ Specifically, hippocampal processing of stored perceptual data and associations is essential to simulation. As Suddendorf and Corballis dramatically put the case (quoted in the previous chapter), "we predominantly stand in the present facing the future rather than looking back to the past" (2007).¹⁷ Schacter and Addis, who pioneered this research, propose the "*constructive episodic simulation hypothesis*" (Schacter and Addis 2007, 2009):

episodic memory provides a source of details for future event simulations, such that past and future events draw on similar information stored in episodic memory and rely on similar cognitive processes during event construction, such as self-referential processing and imagery. Furthermore, we have suggested that the constructive nature of episodic memory supports the flexible recombination of stored details into a coherent simulation of a new event that has not been experienced previously in the same form. This process of flexible recombination is thought to rely on relational processing abilities that are heavily dependent on the hippocampal formation (e.g., Eichenbaum 2001), with recent evidence implicating the anterior hippocampus specifically in recombining episodic details into novel events. (Addis and Schacter 2008)

One implication of this view with regard to event processing is that grasping together details such as imagery and value-for-self into a meaningful time extended sequence (an event) is implicit in processing events as well as reflecting upon and imagining them. Schacter and Addis have hypothesized 'IMAGINING' as a cognitive system that groups together memory, predictive processing, and creative thinking into a common set of interrelated processes.

Several other recent findings about episodic memory processing bear upon event processing. The predictive processing model of episodic memories fits with a view of simultaneous top-down and bottom-up processing.¹⁸ Episodic memories likely shape holistic aspects of present temporal experiences as well as providing sources of details.

Kim (2010) suggests that "many episodic-retrieval-related activations may actually reflect more general attention/executive operations," rather than salience functions (p. 1648). Further, episodic memories provide schematic information that affects how we experience present events. In a research review on sequencing, Davachi and DuBrow (2015) suggest that

the episodic or repeated nature of an experienced sequence may bias the source of temporal information for that sequence: “Temporal information based on a single experience may be biased toward relying on contextual features (e.g., 60 and 61), whereas repetition may bias the source of such information away from a specific spatiotemporal context and toward cued prediction of subsequent items in a sequence” (p. 99). Taken together, these findings suggest that a constructivist view of low-level cognition involves thinking of prior experiences as encoded for flexible reuse in segmenting present experiences into meaningful events.

Findings from Martin Conway offer additional insights into the mutual relation of event processing and episodic memories. Episodic memories “contain summary records of sensory-perceptual-conceptual-affective processing, ... always have a perspective (field or observer), ... represent short time slices of experience, ... are represented on a temporal dimension roughly in order of occurrence, ... make autobiographical remembering specific, ... [and] are recollectively experienced when accessed” (p. 2306).

The temporal structure Conway attributes to episodic memories is similar to events. Both have their parameters and meanings in part determined by other events and organizing schema. Episodic memories are multimodal and have a phenomenology when activated. As ‘summary’ forms of ‘sensory-perceptual-conceptual-affective processing,’ they select and bind experiential details, and these summaries potentially include amodal (linguistic) conceptual information used to make sense of event experiences (p. 2306). The experience of recollection is closely associated with the relation of the event to the subject.

The similarity between events and episodic memories in terms of experiencing may provide a key insight for understanding internal event segmentation. The idea that remembering is experiential may seem surprising. Yet Conway (2005) proposes, contra the generally held view that emotions arise out of interruptions in goal processing, that ‘cognitive feelings’ (such as a sense of closure, accomplishment, and the need to shift attention) may play a role in event segmentation. He further proposes that

many different forms of processing are accompanied by or associated with what we have termed cognitive feelings ... Cognitive feelings allow the individual to experience knowledge. By this view a person not only knows that he or she has just closed the door when leaving the house but also has a feeling, an experience, of a transition from one event to another. (p. 615)

In addition, Conway (2009) proposes that episodic memories have an action to fact semantic structure:

We have proposed that the boundaries of episodic memories are marked at the opening boundary by information about actions and at the closing boundary by facts that are often details about the outcomes of actions (Williams et al., 2008). This is consistent with our more general view that episodic memory and autobiographical knowledge are about goals. That is to say they preserve information that is highly relevant to goal processing including goal generation, plan execution, outcomes and evaluations (for related views see Brewer & Dupree, 1983; Lichtenstein & Brewer, 1980). (p. 2306)

Action to fact patterns suggest that either during or after experiences we attend to relational valuation dynamisms internal to event structures. The experience of an ‘observing perspective’ and cognitive feelings of episodic memories, taken together, further suggest that these dynamic valuations have reference to the self. Research that emphasizes the integration of event selection and episodic memory processes suggests that prior episodic memories facilitate the parsing of present events with these characteristics (action to fact sequences, observing perspectives, and cognitive feelings). In combination with selection for cycles of equilibrium-disequilibrium-equilibrium and action to fact structures based on value dynamism(s), the relation between episodic memory and event processing suggests that our experiences are (often, at least) structured in terms of meaningful temporal strands available for combination and redeployment in other cognitive activities.

6.6 WEAK NARRATIVITY AND EVENT EXPERIENCES

Why hypothesize about a synthetic concept like weak narrativity rather than, as most quantitative work has done, identifying more limited, discrete processes? For some researchers, understanding the cognitive capacities deployed in processing narratives is a key to understanding the simulative function of memory and the Default Mode Network. What, in other words, does our capacity for narrative processing demonstrate about our ordinary cognition?

Schacter, Addis et al. (2012) emphasize the need for further study of narrative processing, particularly its contributions to memory and imagination. Specifically, narrative processing may help explain functions which share a common neural basis, including prospection, scene construction, autobiographical memory, and self-projection (see Spreng, Mar, Kim).

Recall our working definition of narrative processing: ongoing meaning-directed generation and processing of emergent temporal structures consisting of related value-dynamic sequences involving value-dynamic selected agents, mediated by socio-cultural contexts. Narrative processing involves, then, selection, concatenation, determination of time boundaries, determination of changes in agents relative to other agents, their own condition, and selected objects and phenomena. This determination is inseparable from valuing (itself a multi-dimensional or synthetic form of processing).

My hypothesis, developed further in the next chapter (on affect and cognition), is that this synthetic valuation activity is core to ordinary experience. It is in part responsible for selection and boundary definition, as well as mediating socio-cultural pressures, gist processing, and meaning. Indeed, valuation describes ongoing syntheses of these elements that lead to the key elements of Zacks's definition: time segmentation, location, and beginnings and ends.¹⁹ In other words, valuation reflects the subject's ongoing tracking of value dynamics within the continuous flow of experience, and meaning-laden phenomena (like beginnings and endings) as well as more apparently objective phenomena (temporal span and location) emerge from this processing.

This view conflicts with naturalistic assumptions about the "givenness" of certain features of events, as well as how events became familiar; on the other hand, it has certain explanatory advantages. So, is bringing narrativity into event processing parsimonious? Does introducing narrativity create a new, non-empirical speculative layer of processing unnecessarily?²⁰

The case for the necessity of weak narrativity is strengthened by the emergence of predictive processing models of cognition. Once basic cognition is viewed as predictive, we must seek to understand how prediction relates to the anticipatory quality of many of our experiences. Further, once these predictions are viewed as intentionally weighted, we must seek to understand why the time signatures and changes in valuations of selected elements of experience seem freighted with meaning. In the next chapter, I'll hypothesize that the coordination of time signatures and patterns of valuation dynamics includes affectively charged *relational* valuation dynamisms (i.e., valuations of elements that emerge from their relation to other salient elements of experience).

These involve early developing value schema that accommodate temporally extended patterns of moving objects and agents, as well as the observer's subjective, affective response to the event as a whole from an initial state to a concluded state, or as Conway describes episodic memories, from actions to facts (or stable evaluations and/or responses). Taken

together, the event processing and affective dimensions of experience create a sense of weak narrativity as basic to some experiences.

This view seems to be at odds with Zacks' model of event boundaries predicated upon failures of predictions and the instantiation of new predictions. That the experiences drawn upon (episodic memories) themselves have action→fact structures and are themselves viewed as guiding attention and executive functions suggests that attending to and coordinating these relational valuation dynamisms is sensitive to the amplitude of particular dynamisms, which wax and wane until a second equilibrar state is reached or anticipated, and the experience becomes a 'fact' (an outcome). At this point predictions about these relational valuation dynamisms may fail, but equally likely is that they no longer hold interest. Thus, the event segmentation may correspond to the emergence of a new set of predictions that reflects new foci for cognition (it may also, as Zacks theorizes, reflect a failure to understand what's immediately happening, but given the prior success of the predictions, this kind of failure seems counterintuitive).

The explanation of event experiences through predictive processing drawing on episodic memory processes creates a model of present experience in which top-down processing is broader than, though directed by, goals with immediate utility to the subject. Bottom-up information causes a continual recalibration of the predictive response and affective valuing of individual elements and of the whole. A felt experience is dynamic and sensemaking, by which I mean affect normatively contributes to and shapes perception by directing attention and weighting dynamic information in relation to goals (and 'proxy-goals,' as in the example of Theory of Mind in disinterested spectatorship). It is in the context of understanding these dynamics that the definition we've considered of narrative processing begins to correspond to the way we experience the world.

6.7 PHILOSOPHICAL PERSPECTIVES ON WEAK NARRATIVITY IN LOW-LEVEL COGNITION

This chapter has considered some ways in which low-level, or first-order, cognition might involve narrative processing in the chunky messiness of temporal experience, particularly in how events emerge as basic structures of experience. However, as the 'narrativist' positions reviewed in earlier chapters imply, this is a counterintuitive claim: narrative seems most obviously to be a *post hoc*, linguistic configuration of experiences which is

subsequently summarized, synthesized with other narratives and facts, and applied to make sense of novel experiences. Yet there are also signs that we don't feel these *post hoc* acts of configuration sufficiently explain the origins or nature of narrative. Take, for example, philosopher Judith Butler's description of this configuration: "My narrative begins *in medias res*, when many things have already taken place to make me and my story possible in language. I am always recuperating, reconstructing, and I am left to fictionalize and fabulate origins I cannot know" (p. 39). For Butler, narrative *enables* story but is not coextensive with it.

Richard Menary, one of the leading thinkers challenging top-down views of how narrative might affect cognition, offers the following observation about how narrative structures pre-exist narrative configuration:

It is not narratives that shape experiences but, rather, experiences that structure narratives. Experiences are the sequence of events that give structure and content to narratives. There may be additions and elaborations to this embodied sequence at a later time, after reflection, but the temporal ordering, the structure is already there in our lived, bodily experience. The mistake is again to suppose that a narrative conceived in abstraction could be brought to bear on a sequence of experiences, ordering them and giving them meaning. (p. 79)

From this perspective, the autobiographical approach to narrative as part of cognition (see Chap. 2, Sect. 2.4) is difficult to square with a predictive processing model and with recent research on the nature of episodic memory. While notions like the 'teller effect' will be considered in the final chapter, autobiographical mental work does not provide a good model for understanding the dynamics in mental action that cause episodic memories to form. Menary's view is that experiences "are ready to be exploited in a narrative of those experiences" (p. 75).²¹

As this chapter (and the next) describe, sequential associations inherently reflect different types of selection, which are likely generated through multiple distributed processes. What Menary calls 'pre-narrative fodder' may be highly structured, selected for and embedded with a variety of elements, including:

- for-me-ness (subjective perspective, self-not-self boundary definition);
- event segmentation (temporal boundaries);

- sensemaking (i.e., association of dynamics, objects, other phenomena to varied causal and analogical phenomena (categorization, intention-reading, prediction, etc.);
- relevance (active attention, signal/noise, and equilibrar monitoring functions);
- valence (affect);
- concatenation (forming temporally based associative sequences that integrate various elements, including ‘unlike’ elements, via varied criteria including causation).

These elements contribute to experiences ripe with sequential associations characterized by perspective, segmentation, causality, analogy, rich temporality (including memory and prediction), relevance, and valence.

We’ve considered theories of event processing in some detail in this chapter. The idea, simplified, is threefold: weak narrativity could exist in linkages between events, in the internal features of events, or in the processes that shape experience into events. First, if meaningfully linked events comprise narratives, and our basic experience is unconsciously and consciously ‘chunked’ into events, there is a possibility of weak narrativity in experience. Also, if event experiences are dependent on internal features that make them internally narrative or ripe for meaningful linkage to other events, there is a possibility of weak narrativity in experience. Third, the processes behind event definition include selection, boundary delimitation, and internal dynamism that accounts for coherence and meaning while parsing condition changes. If these processes are based upon making sense of the present via meaningful temporal and relational patterns, and if they result in experiences that are shaped according to these patterns, and if these patterns have recognizably narrative elements (such as equilibrium-disequilibrium-equilibrium structure and action→fact structure), then there is a weak narrativity in experience.

What is rich about event recognition that is not adequately captured in current models of events? First, a narrative logic may be more useful, or at least a crucial supplement, to action-orientation logics when making sense of events that the subject has no particular stake in, including events whose demand for attention exists outside of any immediate goals other than interest. Simply, the fuzziness of events points to their inextricability from ongoing processing, which includes the ‘sentinel’ functions, higher order cognitions (including reflection, confabulation, and the like), and other event processing. That is to say, event processing is never fully discrete,

and the location of individual event processing within an ecology of sense-making and higher order functions points to a linking process between events that draws upon semantically laden predictions. To describe this messy ecology, the language of 'script' seems inadequate because a script seems directed toward discrete, unrelated events. To invoke narrativity, then, is to suggest that at a basic level of experience our sensemaking and anticipatory faculties are continually engaged in event processing activities that emerge from and are re-sublimated within this messy ecology via linkages between experiences, present, remembered, and anticipated, that are essential to the dynamics of event selection, boundary definition, and internal parsing. Sense doesn't emerge from non-sense immediately. Instead, it emerges from a messy ecology of partial, inchoate, iteratively parsed, ongoing sensemaking. Scripts, event schemata, and other sense-making cognitive resources likely are key to the emergence of sense from this messy ecology.

Our experience of an event is a defining of the event in deep relation to like and proximate events. Events exist in a half-light, generated out of the mess of experience in relations of value to other events and features of experience, and gaining fuller definition under the scrutiny of reflection and communication. Event definition is a processing that emerges and fails continually. An event is defined as singular, and thus detachable from the flow of experience and rememberable as a model for future experience once it achieves a certain sense of stable structure, as indicated by the equilibrium-disequilibrium-equilibrium and action→fact vectors described above. Once 'stabilized' (see above for commentary on event and episodic memory stability), the traces of its roots in the messy, and possibly narrative, ecology of experience are invisible because they have been built into the structure of the event. The event-object, as far as it exists, is structured by its linkages to other events.

Events are also deeply conditioned by the kind of attention higher order concepts cause us to pay. Take the example of sitting in your seat when a woman standing before you points a gun directly at her head. If you are not expecting this, it is sure to provoke a visceral response.

However, if you are not expecting it but you are watching a play and you believe the woman is an actor, the visceral nature of the response is muted. And of course if you are expecting it—say you've seen the production before—you are unlikely to have much of a visceral response at all. That is, our immediate experience is deeply conditioned by 'higher order,' conditional understandings.

The messy ecology/narrative processing view exists in some contrast to, on the one hand, Richard Menary's view; and on the other, to the view expressed in Zacks et al. Note that these two views themselves are challenges to classical models, of narrative in experience on the one hand (see Chap. 2, Sect. 2.3 and Chap. 3, Sect. 3.3), and of cognition and event processing on the other.

In concluding this chapter I want to return to the premise of this book: to argue that weak narrativity is generated in cognitions of ordinary experiences, and to help dispel some unhelpful ideas about narrative and cognition in light of emerging contemporary research into how certain features of cognition operate. Philosophers have long posited that experience is meaningful—that is to say, conscious experience comes preprocessed with meanings, rather than raw. Normative conscious experience involves navigating meaning-laden environs, relations, and situations, rather than taking meaningless percepts and deriving or weaving meanings out of them.

The hermeneutic tradition foregrounds meaning in experience, emphasizing the interpretive nature of our interface with the world. As Meretoja (2013) notes, “in the Heideggerian-Gadamerian [tradition of] hermeneutics ... understanding is the fundamental structure of our being in the world,... all experience has the structure of interpretation, of ‘understanding something as somethings’” (p. 99). Hermeneutics offers several descriptions of experience that resonate with the narrative features I am describing. First, it distinguishes itself from the phenomenological tradition, which leaves open the possibility of naive (i.e., unprocessed) experience. The hermeneutic perspective, by contrast, views experience as a form of interpretive processing. Interpretation is in this view ontologically significant, in addition to being an epistemological tool (Meretoja 2013, p. 93). Views of narrative within hermeneutic analyses are similarly themed and similarly varied: narrative is seen as fundamental to interpretation, as a kind of commentary, as a “human way of shaping experiences into a meaningful temporal continuum” (Meretoja 2013, p. 103), and as crucial to meaning and ethical reflection. Within conversations in hermeneutics, the makeup and function of narrative vary along axes including representation/epistemology, action/ontology, instrumentality/ experientiality, embodied/mental, hindsight/immediacy, and self-focused/ outwardly focused. A central tenet of Gadamer's hermeneutics is ‘linguisticity,’ or the necessity of language for interpretation.

Paul Ricœur is a foundational figure in linking hermeneutics to narrative. Working in the phenomenological and hermeneutical traditions,

Ricœur views temporal experiences as thick with an evolving present suffused with the impacts of previous moments and with anticipation of future instances. Ricœur's argument, in the magisterial *Time and Narrative* and subsequently in *Oneself as Another*, points to the importance of a language-based, reflective narrative identity that develops in response to lived experience. Many of Ricœur's readers have concluded that Ricœur sees a discontinuity between narrative and lived experience. Yet according to Lewis, Ricœur is not fully committed "to the notion that lived experience is devoid of narrative structure" (p. 13). Instead, both lived experience and its narrative structuration are subject "to the dominance of mediation" (Lewis 13).

As the research reviewed earlier on episodic memory suggested, episodic memory shares experiential and processing bases with event segmentation. It includes multimodal (non-linguistic) inputs, but may be linguistically mediated as well. It also specifies autobiographical memory and contributes to our ongoing, continuous feeling of future-orientedness. As such, the episodic memory/event segmentation relationship seems fundamental to empirical support for a hermeneutic perspective. And as we've seen, narrativity—but not artifactual narrative—is plausibly component to each. Viewing weak narrativity as a way to describe Ricœurian mediation (via the production of temporal strands with semantic features) helps explain how narratives refined in higher level thinking could ever impact our immediate experiences or even the concepts and values that guide those experiences. For our concepts and values are products of iterative learning, which involves processes of abstraction from experience such that habitual lived experiences are increasingly mediated by perspective.

Ultimately, what's at stake is an effort to more richly describe the cognitive dynamics that produce experience. Conceptually, weak narrativity describes event segmentation and episodic memory in terms of specific, sensory-perceptual-cognitive-affective details (some of which are dynamic) and self-reference, and a tendency toward 'condition→action→fact/valuation' sequencing. Conceived in this way, experiential processing likely to result in episodic memory reflects a dynamic associative process of potentially 'unlike' details for defining event markers that are meaningful. By meaningful, I mean they (a) are employed to construct a form that none of the detail elements represents on its own; (b) regulate the shift from Default Network processing to active attention and back again; (c) employ valuation processes drawn from affective, sensory, and cognitive facilities that mediate self-reference; (d) result in episodic memories structured as 'condition→action→fact/

evaluation' sequences. Together, I've described these aspects of lower order cognition as constitutive of basic 'narrative' processing. It is suitable for more complex narrative processing in which social and personal narratives/narrative-embedded valuations might figure. In the next chapter, we'll consider the role affect plays in making experience 'hot' and meaningful.

NOTES

1. The shape of temporality depends on a combination of perceived social construction, individual intentional behavior, and unconscious processes. There is not space here to review the range of debates about this issue, particularly in process philosophy, but along with James, John Dewey and Henri Bergson offered divergent perspectives on the nature of structure in time and its relation to the felt experience of meaning.
2. Events grasped in presence also reflect semantic memory. For example, a savvy health practitioner can quickly grasp the onset of anaphylaxis in a patient allergic to latex. In this case, immediate grasping of events depends upon deep knowledge: Her knowledge of the patient, and her knowledge of the symptoms, triggers, and progression of anaphylactic shock, are inseparable, in all likelihood, from regimes of more and less optimal intervention (reflecting the doctor's goal of treating the patient with minimum risk of harm and maximum optimization of the patient's immediate and long-term well-being).
3. It seems highly likely that the shift in attention is not so clean, moving from one locus of activity to a new set of environmental circumstances. Instead, it seems likely that 'attending' becomes diffuse, moving from a near-total focus on the dynamics of the falling tree to a partial attention to settling noises accompanied by gradual awareness of the new environmental circumstances the event has created.
4. The notion of a 'disruption' of order may not seem to apply to the 'getting ready for work' example. However, we might think of transitions between goal-directed activities as involving the interruption of a goal upon whatever else occupies the subject's cognition, followed by the pursuit of the goal via the various subordinate activities described earlier, to resolution, as if the goal is 'checked off' on one's to-do list.
5. This is a primary way in which the influence of sociocultural values impacts low-level event processing.
6. Hommel updates the theory in 2019 to claim that representation may not be central: "recent theoretical developments provide the basis for a more

integrated view consisting of both the codes that are shared between perception and action in the control processes operating on these codes. Four developments are discussed in more detail: The degree to which the integration and retrieval of event files depends on current goals, how metacontrol states impact the handling of event files, how feature binding relates to event learning, and how the integration of non-social events relates to the integration of social events” (p. 2139).

7. See the examples above for a sense of what this might entail; also, consider the commentary on value in Chap. 7.
8. This will be counterintuitive for readers who think of, say, reading a novel as an apt illustration of experiencing narrativity. The reader opens a container that bounds the narrative, and maybe encounters an internal container (think of framing narrators and the distinction between story and discourse). The dynamic elements are set into motion by elements at the boundaries. If we want to understand narrative features of experience (or, more simply, if we believe experience has a narrative character), however, we have to grapple with how false the ‘reading a novel’ analogy is. Instead, reading a novel might be best understood as a way to encounter narrativity in experience. It is not itself a narrative experience, but an experience of a highly wrought narrative. At the same time, ‘inside-out’ and ‘container’ theories are not logically diametrically opposed—that is, structure could emerge from both the inside-out and imposed external scripts simultaneously.
9. Here I am not addressing the retrospective configuration of many events. For sports fans, consider when a team goes on a ‘run’—that is, it has an extended string of successes. If this composite of events is itself an event, then the recognition of the event is *post hoc*. This is related to Meretoja’s second level of interpretation.
10. Bar’s theory depends on rapid ‘gist’ processing of essential details from a situation. For more on ‘gist’ processing, see Chap. 8, Sect. 8.4.
11. In this view, ‘what it is like’—that is, the phenomenology of an experience—is a synthesis of diverse intentional weightings, perceptual selections, perceptual capture, and cognitive processing.
12. See Chap. 3 for Herman’s ‘gradient’ conception of narrative and Currie’s notion of narrativity.
13. EST claims that “event segmentation is a spontaneous concomitant of ongoing perception, happens simultaneously on multiple time scales, and depends on change” (15).

14. To effectively predict the unfolding of ordinary events, event models “need a particular type of dynamics: stability punctuated by phasic transitions [which]...segment ongoing experience into a succession of events represented by successive working models” (966).
15. The research on event segmentation opens up/creates boundary spaces for fill-in explanations, and that these spaces are where weak narrativity, if it exists, is likeliest located. Weak narrativity makes sense if it describes dynamics necessary for event processing to proceed or be utilized.
16. Hill and Emery argue that “Working Memory contributes to the construction of a single, coherent, future event depiction, but not to the retrieval or elaboration of event details” (p. 677).
17. See also Mullally and Maguire 2013 “Memory, Imagination, and Predicting the Future.”
18. 185. For example, Tsuda et al. (2015) find that “a constructed memory itself does not represent an input; rather it represents input/output relationship via bifurcations. The associated dynamics was chaotic itinerancy” (Tsuda et al. 2015).
19. Beginnings and endings are an important consideration—for EST, they reflect the breakdown of predictions and the installation of new models. In narrative, beginnings and endings reflect the initiation and completion of sequences with narrativity. This reflects EST’s focus on continuous experience, and the event as a structuring of that experience. In narrative, beginning and ending are oriented toward the continuity of the sequence as a coherent whole. The question for EST, and where ‘weak’ narrativity comes in to play, is whether the orientation toward structuring continuous, flowing experience can really focus on the ‘dynamism’ or ‘disruption’ that causes a breakdown in prediction. To accommodate dynamism-in-continuity, the event segmenting approach must involve some metrics of anticipated dynamisms (appropriate-to-the-event dynamics, vs. not appropriate). This means that anticipation of change must itself entail expected sequences of change. These ‘ends’ undoubtedly encode information about agential goals and their realization as well as about the emergence of static conditions after dynamic conditions (whether physically, relationally, or otherwise). So, information about endings may be part of predictive processing, as well as cued by disruptions.
20. An alternative might be simply to stick with ‘episodicity.’ However, thinking only of episodicity misses how episodicity can be emergent in the context of enchainment of episodes. That is, the episode definition is conditioned by ongoing weakly narrative event concatenation.
21. Menary is a realist about experiences, rather than adopting the predictive processing and affective constructivism considered in this and the next chapter.

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